

Annex — How to read this report

Introduction

This appendix accompanies the Portfolio rotation report and presents the technical, methodological, and interpretive foundations of the model. It describes the structure of the data files, the functioning of dynamic windows, the calculation formulas of key indicators, and the interpretive logic used in the tables and scenarios of the main report. The objective is to ensure methodological transparency, reproducibility of results, and a clear understanding of the signals generated by the program, regardless of the asset type or time horizon considered.

1. Objective & Scope

The purpose of this report is to **describe and interpret the state of a given asset universe**, whether equities, indices, ETFs, or other instruments. It aims to **identify phases of rotation, tension, or stabilization** through the combined analysis of returns, volatility, momentum, and their coherence across short- and long-term horizons. The report therefore provides a **synthetic, structured, and interpretable overview** of market regimes, composition biases, and risk conditions, facilitating **global diagnosis and decision-making**.

- Sources: *sectors.parquet*, *vol_profiles.parquet*, *constituents.csv*.

sectors.parquet: The *sectors.parquet* file contains the historical prices of the assets used in the analysis. The program uses it to compute returns, volatility, and momentum over different time windows (short- and long-term). In practice, this is the primary database from which all indicators, correlations, and rankings presented in the report are derived.

vol_profiles.parquet: The *vol_profiles.parquet* file contains structural and qualitative information for each asset: volatility profile, market regime, liquidity, leverage, beta, asset class, market capitalization, etc. These data are not derived from time-series calculations but from prior statistical profiling or long-term structural analysis. Complementing *sectors.parquet*, it provides a macro and contextual interpretation layer useful for understanding the expected behavior of each asset (e.g., “dynamic,” “speculative,” “defensive”). The two files are kept separate to distinguish raw data (price history) from descriptive metadata (risk profile). This separation makes the model more modular, readable, and reusable while avoiding unnecessary recalculation or duplication of information during market data updates.

constituents.csv: The *constituents.csv* file contains the reference list of analyzed assets, including their symbols, full names, and GICS classifications (sector, industry, etc.). It does not provide market data or computed metrics, but rather a descriptive mapping that identifies each entry in *sectors.parquet* and links tickers to their corresponding economic or thematic sector. This file therefore acts as a bridge between numerical data and their economic context, essential for generating comments on diversification, sector bias, and macro coverage. It is kept separate to maintain a clear and easily updatable structure (e.g., when a company changes sector or ticker symbol without affecting its historical series).

WARNING — *constituents.csv* file: This file was created manually and is not automatically updated. Before running the program, the user must verify that all assets included in the analysis are properly listed in this file with the correct symbol and classification (GICS sector, sub-industry, etc.). If any assets are missing, they must be added manually to ensure the consistency of the report and the correct generation of sector-related comments.

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2. Windows & Frequency

2.1 Dynamic Windows.

Dynamic windows are a central component of the project: they allow the analysis periods (volatility, return, momentum) to automatically adjust to the dataset's frequency and depth.

2.2 Purpose

The goal is to avoid a fixed calibration (e.g., 20 days for everything), which would be inconsistent when switching between daily, weekly, or monthly data. The program instead computes window sizes proportional to the total number of available observations, ensuring that indicators remain comparable and economically coherent across all horizons.

2.3 Functioning

1. The data frequency is first detected automatically (Daily, Weekly, Monthly, etc.).
2. Based on this frequency, a base window size is set (e.g., 20 for daily, 4 for weekly, 3 for monthly).
3. This base is then dynamically adjusted according to the total number of data points:
 - Short-term (CT) windows typically cover about 10% to 30% of the historical series.
 - Long-term (LT) windows extend up to roughly 85% to 90% of the series.
4. Safeguards prevent the use of excessively short or long windows (minimum 3 points, maximum $n-5$).

2.4 Distinct Windows per Indicator.

- Volatility: requires a shorter, more reactive window since it measures the instantaneous dispersion of returns.
- Return: uses an intermediate window, representative of the average behavior over a meaningful period.
- Momentum: requires a longer window, as it reflects cumulative trends and must avoid overly noisy signals.

WARNING — For datasets that are too short, dynamic windows mechanically become too small, increasing the volatility of computed indicators. They do not replace expert economic judgment: 'optimal' windows may vary depending on the nature of the assets (e.g., crypto vs. bonds). Finally, while the dynamic approach ensures structural robustness, it limits cross-study comparability if the underlying data depth differs significantly.

3. Portfolio Structure & Macroeconomic Profile

3.1 Purpose

The *Portfolio Composition* section aims to present the initial structure of the analyzed portfolio by identifying its sector allocation, capitalization biases, and main areas of concentration. It provides a static and macroeconomic context from which subsequent dynamics (returns, volatility, regimes) will be interpreted.

3.2 Functioning

The *Portfolio Composition* block relies on metadata from *constituents.csv* and structural profiles from *vol_profiles.parquet*. Each portfolio line is mapped to a sectoral category (GICS), then aggregated as a relative weight, either:

- equally weighted (weight = $1/n$),
- or weighted by market capitalization when available.

3.3 Analytical usefulness

Before analyzing performance or signals, it is essential to understand what the portfolio is invested in sectors, defensive biases, or overweight exposures. Indeed, a portfolio highly concentrated in three or four sectors, or dominated by mega-caps, does not react the same way as a balanced basket. This section allows assessment of whether the observed market dynamics (momentum, volatility, rotation) stem from the portfolio's construction or from external macro factors.

3.4 Functioning of the Macro Profile Insight Module

The automatic *Macro Profile Insight* commentary relies on a multi-level analytical logic combining descriptive (sectoral) information with structural (quantitative) profiles to produce a coherent synthesis of the portfolio's stance. This interpretative engine operates from two main datasets:

- *constituents.csv*: GICS classification and nominal composition of the portfolio;
- *vol_profiles.parquet*: structural metadata (volatility, beta, liquidity, market capitalization, leverage, asset type, etc.).

The module first analyzes sector diversification based on GICS labels. The number of distinct sectors determines the degree of macroeconomic coverage:

- A single sector → extreme concentration, dependency on a single macro driver;
- Three to five sectors → partial coverage, strong thematic bias;
- Eight to ten sectors → balanced diversification, close to a global benchmark.

Absent sectors are then identified to infer implicit biases:

- Absence of Energy or Commodities → lack of inflation protection;
- Absence of Healthcare or Consumer Staples → weak defensive component;
- Absence of Technology or Communication → anti-growth bias.

The engine then analyzes structural profiles from *vol_profiles.parquet* to complete the interpretation:

- **Market Cap Label:** detects mega/large-cap tilts (stability, low idiosyncrasy) or small/micro-cap tilts (higher convexity, increased volatility).
- **Asset Type:** measures the proportion of ETFs, REITs, or ADRs to identify the nature of exposure (indirect, real estate, geographic).
- **Liquidity Label & ADV10USD:** assesses the liquidity stance (e.g., ample trading liquidity, pockets of thin/illiquid names).
- **Beta Label & Leverage Label:** describes systematic sensitivity and any leverage effects present.
- **Volatility Profile:** defines the portfolio's overall style (balanced, defensive, or speculative).

The purpose of *Macro Profile Insight* is not to assess performance but to characterize the portfolio's structural stance prior to any dynamic interpretation.

WARNING — The information presented in this section is based on the portfolio composition as declared at the time of analysis and on categorizations from *constituents.csv* and *vol_profiles.parquet*. It does not necessarily reflect real-time weights, indirect economic exposures, or recent portfolio adjustments.

4. Key Indicators

This section presents the core indicators used in the model, which are Return, Volatility, Momentum, Risk-Adjusted Return, Signal Stability, Volatility Regime, and ADV10USD. Each captures a specific dimension of market dynamics: performance, risk, consistency, or liquidity. Together, they form the analytical foundation on which the report's interpretations and scenarios are built.

4.1 Return / R (%)

Return measures the average performance of an asset over a given period. It is the foundation of all market analysis, indicating both direction and intensity of price movement. It enables time-normalized comparison of assets or sectors regardless of volatility or momentum levels.

The average return (AvgReturn %) is computed over a dynamic rolling window whose size depends on the detected data frequency. This ensures that calculation sensitivity adapts to historical depth and typical market volatility.

- The program first computes logarithmic returns between two consecutive observations (making variations additive over time).
- These returns are then aggregated over a moving window; its size varies by frequency (≈ 20 days for daily, ≈ 4 weeks for weekly, etc.). The longer the frequency, the wider the window, smoothing noise while capturing dominant trends.
- The total return over the window is converted into an average percentage per period, providing an intuitive measure of how much the asset gains or loses on average per observation period.

Return interpretation by horizon ()*

Horizon	Sharp rise	Moderate rise	Neutral	Mild drop	Sharp fall
Daily	> +1.5%	+0.8 → +1.5%	-0.2 → +0.2%	-1.0 → -0.2%	< -1.0%
Weekly	> +4.0%	+2.0 → +4.0%	-0.5 → +0.5%	-2.0 → -0.5%	< -2.0%
Monthly	> +8.0%	+4.0 → +8.0%	-1.0 → +1.0%	-5.0 → -1.0%	< -5.0%
Yearly	> +25.0%	+15.0 → +25.0%	-5.0 → +5.0%	-15.0 → -5.0%	< -15.0%

Return is the report's pivot: it feeds the other indicators (momentum, ratios, scenarios), frames the economic reading of cycles (rally, correction, rebound, etc.), and serves as a comparative benchmark in identifying stress phases or sector rotation. Without Return, directionality and consistency analysis would be impossible.

WARNING — Return says nothing about risk: a high return may come with high volatility. It can be misleading over short horizons, especially after one-off events (earnings, news, etc.). It does not reflect persistence: frequent back-and-forth swings can yield a near-zero average return even in a very active market. The mean return is not annualized, as it captures local dynamics, not one-year performance.

4.2 Volatility / V (%)

Volatility measures the average amplitude of price fluctuations over a given period. It is an indicator of risk and instability, complementary to return: while Return shows direction, Volatility reflects nervousness. It helps assess an asset's regularity, shock sensitivity, and ability to sustain a stable trend. In this report, volatility plays a central role in interpreting market regimes (subdued, normal, elevated, turbulent).

Volatility is computed using the same dynamic rolling-window logic as Return and Momentum:

- The program computes the statistical dispersion (standard deviation) of logarithmic returns over a moving window.
- Window length depends on analysis frequency: the higher the frequency, the shorter the window (≈ 20 days daily, ≈ 4 weeks weekly, ≈ 3 months monthly, etc.).
- The result is expressed as an average percentage per period, providing comparable scales across assets and horizons.

This dynamic approach keeps short-term volatility reactive while preserving long-term structural readability. It adapts automatically to available historical depth, preventing arbitrary fixed windows from distorting comparability across assets.

Volatility interpretation by horizon ()*

Horizon	Subdued	Normal	Elevated	Turbulent
Daily	< 0.3%	0.3 – 0.8%	0.8 – 1.5%	> 1.5%
Weekly	< 1.0%	1.0 – 2.5%	2.5 – 4.5%	> 4.5%
Monthly	< 3.0%	3.0 – 6.0%	6.0 – 10.0%	> 10.0%
Yearly	< 8.0%	8.0 – 15.0%	15.0 – 25.0%	> 25.0%

Volatility identifies the overall risk regime in which the portfolio operates; it determines the likelihood of trend change, momentum stability, and short-term signal coherence. It also underpins Risk-Adjusted Return and drives the classification of market scenarios (capitulation, stress, uptrend, etc.).

WARNING — Volatility measures neither direction nor future performance: a bullish market can be highly volatile, and a bearish one can be stable. A temporary decline in volatility does not always mean reduced risk, and it may precede a breakout. Over short series, statistical instability may occur, especially when returns show jumps or irregular extremes.

4.3 Momentum / M (%)

Momentum measures the directional persistence of price movement over a given period. While Return captures average performance, Momentum describes the coherence and speed of the move whether the market advances steadily or oscillates indecisively. It is key to detecting acceleration, exhaustion, or trend reversal phases.

Momentum (M %) is computed over a dynamic rolling window adjusted to data frequency:

- When the window has 4 points or fewer (typically daily / weekly), sample size is too small for a meaningful average. In this case, the program sums the logarithmic returns, a raw accumulation of consecutive variations. This keeps the signal responsive: on very short horizons, each individual change carries weight, and normalizing by w would dampen true amplitude.
- When the window exceeds 4 points, raw summation loses relevance as returns offset each other. The program then uses the geometric mean of logarithmic returns, expressing the average rate of price progression per time unit while neutralizing window-length effects.
- The result is normalized by window size to yield an average per time unit.

This hybrid approach preserves consistency of momentum across horizons without sacrificing precision on short ones.

Momentum interpretation by horizon (*)

Horizon	Weak	Moderate	Strong	Extreme
Daily	< 0.5%	0.5 – 1.5%	1.5 – 3.0%	> 3.0%
Weekly	< 1.5%	1.5 – 4.0%	4.0 – 8.0%	> 8.0%
Monthly	< 3.0%	3.0 – 6.0%	6.0 – 12.0%	> 12.0%
Yearly	< 8.0%	8.0 – 15.0%	15.0 – 30.0%	> 30.0%

Momentum complements Return: it indicates not absolute direction but relative strength. An asset may show positive return yet weak momentum (trend fatigue) or negative return with improving momentum (technical rebound).

WARNING — Momentum is timing-sensitive: too short a window amplifies noise; too long a window hides fresh signals. It ignores volatility level and a fast move may reflect panic as much as conviction. A very high momentum is not automatically bullish: it can mean overheating or imminent reversal. Momentum should therefore always be read together with Volatility and Return to confirm consistency or detect divergences.

4.4 Risk-Adjusted Return / RAR

The Risk-Adjusted Return (RAR) measures average performance relative to risk level, represented here by volatility. It is a simplified, intuitive Sharpe-ratio-type metric assessing efficiency: how much return an asset generates per unit of volatility. It enables comparison across very different profiles on a homogeneous basis. A high RAR means strong compensation for risk; a low or negative RAR implies inefficient or risky performance.

In the program, RAR is computed as:

$$RAR = AvgReturn (\%) / Volatility (\%)$$

- Both values use the same dynamic windows as the base indicators.
- It is not annualized and it reflects instantaneous efficiency rather than long-term performance ratio.
- The result is unit-free (pure ratio) and centered around 0.

Risk-Adjusted Return interpretation

Horizon	Unfavorable	Neutral	Efficient	Exceptional
Daily	< 0.5	0.5 – 1.0	1.0 – 2.0	> 2.0
Weekly	< 0.5	0.5 – 1.0	1.0 – 2.0	> 2.0
Monthly	< 0.4	0.4 – 0.8	0.8 – 1.5	> 1.5
Yearly	< 0.3	0.3 – 0.6	0.6 – 1.2	> 1.2

RAR provides a direct reading of return-to-risk balance, automatically adapting to frequency and data depth. It is independent of absolute return scale: it expresses quality rather than quantity.

WARNING — RAR ignores correlations with other assets (it is not a portfolio metric), is insensitive to asymmetric or fat-tailed distributions, and can be misleading in low-volatility phases when performance is temporarily high. An asset with high β may show strong RAR simply due to market amplification. For a fairer view, interpret RAR in light of beta sensitivity (β) and the asset's volatility profile.

4.5 Signal Stability

Signal Stability measures the consistency between short-term momentum and long-term momentum. It indicates to what extent recent dynamics confirm or contradict the dominant market direction. This indicator qualifies the reliability of the ongoing move whether the signal is coherent, fragile, or reversing.

In the program, Signal Stability is computed as:

$$\text{Signal} = (\text{Momentum} \times \text{Momentum_LT}) / (|\text{Momentum}| + |\text{Momentum_LT}| + \varepsilon)$$

(with $\varepsilon \approx 10\text{e-}6$ to avoid division by zero)

The result is a normalized index between -1 and $+1$:

- $+1$ → perfect coherence (short- and long-term momentum aligned)
- 0 → no clear relation or weak contradictory signals
- -1 → full divergence (strong trend reversal)

Signal interpretation

Value	Signal state	Interpretation
$> +0.6$	Perfectly aligned / Strong coherence	Short-term momentum fully confirms the long-term trend. Robust and reliable signal.
$+0.2$ to $+0.6$	Partially aligned / Moderate coherence	The direction remains broadly aligned, but the signal shows some loss of strength.
-0.2 to $+0.2$	Neutral / Indeterminate	No clear trend: possible sideways phase or pause before a new movement.
-0.6 to -0.2	Emerging reversal / Early turning point	Early stage of reversal: short-term momentum starts opposing the underlying trend.
< -0.6	Strong reversal / Confirmed inversion	Clear inversion: reversal confirmed, often seen in correction or capitulation phases.

This signal helps detect market phase transitions (bull → bear or vice versa), validates momentum / return consistency, and anchors scenario classification (coherence, distribution, stress, etc.).

WARNING — May become unstable on very short series (insufficient points for LT momentum), cannot distinguish volatile spikes from genuine reversals (cross-check with volatility), and its $[-1, +1]$ range is qualitative rather than probabilistic, it is structural, not predictive.

4.6 Volatility Regime

The Regime reflects the market's overall state through the level and dynamics of volatility. It is a qualitative typology describing the risk environment in which the asset or portfolio evolves (calm, normal, tense, or chaotic). Its purpose is to provide an intuitive reading of dominant volatility phases rather than a raw numeric value.

The Regime is derived from *vol_profiles.parquet*, which aggregates the historical analysis of each asset's average volatility. Each record contains structural information used to classify its profile.

Computation of Regime relies on the ratio between two volatility horizons:

- *vol_short* – volatility on the short-term window (e.g. 20 days daily, 4 weeks weekly);
- *vol_long* – smoothed volatility on the long window (e.g. 6–10 × longer).

The resulting ratio produces automatic classification into volatility profiles (defensive, balanced, dynamic, speculative).

Volatility regime interpretation

Regime	Brief Description	Market Meaning
Subdued	Abnormally low volatility, calm market	Consolidation phase, complacency, or waiting period. Potential breakout risk.
Normal	Volatility aligned with historical average	Balanced market conditions, stability, and clear visibility.
Elevated	Above-average volatility	Rising nervousness and uncertainty — tension or mild stress phase.
Turbulent	Extremely high volatility, severe instability	Panic, market disorder, systemic stress, or capitulation.

This signal forms the foundation for dynamic scenario analysis, weighting the other indicators (Return, Momentum, Signal) according to the volatility context and helping detect regime shifts often preceding major reversals.

WARNING — The Regime is descriptive, not predictive: it captures current risk conditions, not future changes. It depends on the quality of historical profiling (*vol_profiles.parquet*). A short or biased sample may distort classification. Structurally volatile assets (e.g. crypto, small caps) may stay in “elevated” for extended periods without being abnormal. Regime transitions may lag due to window smoothing.

4.7 ADV10USD (Average Daily Dollar Volume – 10 days)

ADV10USD represents the average dollar amount traded daily over the last ten sessions. In other words, it measures an asset’s operational liquidity and its ability to absorb buy / sell volumes without causing excessive price impact.

This field is not computed by the program; it is retrieved from Yahoo Finance and stored in *vol_profiles.parquet*. For non-USD listings, the provider converts or exposes an equivalent in USD using the current FX rate.

ADV10USD interpretation

Market Cap (indicatif)	very_liquid	liquid	market	thin	illiquid
Mega (>\$200B)	≥ \$200M	\$75–200M	\$30–75M	\$10–30M	< \$10M
Large (\$10–200B)	≥ \$100M	\$40–100M	\$15–40M	\$5–15M	< \$5M
Mid (\$2–10B)	≥ \$50M	\$20–50M	\$8–20M	\$3–8M	< \$3M
Small (\$300M–\$2B)	≥ \$25M	\$10–25M	\$4–10M	\$1–4M	< \$1M
Micro (<\$300M)	≥ \$10M	\$4–10M	\$1.5–4M	\$0.5–1.5M	< \$0.5M

It is a key indicator for assessing market depth and execution risk. The higher the ADV, the more liquid the asset with large orders can be executed with limited impact. Conversely, a low ADV indicates a thin market where each transaction influences price more heavily.

WARNING — This metric does not reflect long-term structural liquidity: it is sensitive to recent events (earnings, macro announcements, sector news) and should be interpreted cautiously in quiet or atypical periods. ADV10 covers only ten sessions (~ two weeks of trading). It is highly reactive but also unstable, as a single exceptional volume day (e.g. earnings release or merger news) can inflate the average and create a false impression of lasting liquidity.

5. Short-Term vs Long-Term Comparison

This section compares short-term (CT) and long-term (LT) dynamics to evaluate the coherence or divergence of market signals. The goal is to determine whether an asset is in a phase of acceleration, stabilization, or reversal.

The indicators used here (ΔReturn (pp), Momentum Ratio, Volatility Ratio, and Signal Stability) are relative measures: they normalize CT values against LT values. As a result, they are independent of absolute time units or asset types and can be compared across equities, indices, commodities, or currencies.

CT↔LT coherence indicates a clear structural trend, while prolonged divergence may reveal sector rotation, cycle exhaustion, or the onset of a correction.

5.1 ΔReturn (pp) — Return Variation

The ΔReturn (pp) variable (delta return, in percentage points) measures the gap between an asset's short-term and long-term average returns. In other words, it captures the recent acceleration or slowdown in performance relative to the broader trend.

It is a differential trend indicator:

- A positive ΔReturn indicates improved recent performance versus the long-term regime (price acceleration, rebound, or rally).
- A negative ΔReturn indicates deterioration or loss of momentum (correction, pullback, consolidation).

The formula is straightforward:

$$\Delta\text{Return (pp)} = \text{AvgReturn}(\%) \{CT\} - \text{AvgReturn}(\%) \{LT\}$$

That is, the difference between short-term and long-term average returns, each computed over distinct dynamic windows defined earlier in the report.

ΔReturn interpretation

Horizon	ΔReturn (pp) — Strongly negative	ΔReturn (pp) — Moderately negative	ΔReturn (pp) — Stable	ΔReturn (pp) — Moderately positive	ΔReturn (pp) — Strongly positive
Daily	< -2 <i>Sharp deterioration</i>	-2 to -0.5 <i>Weakening</i>	-0.5 to +0.5 <i>Stable</i>	+0.5 to +2 <i>Acceleration</i>	> +2 <i>Strong acceleration</i>
Weekly	< -1 <i>Sharp deterioration</i>	-1 to -0.25 <i>Weakening</i>	-0.25 to +0.25 <i>Stable</i>	+0.25 to +1 <i>Acceleration</i>	> +1 <i>Strong acceleration</i>
Monthly	< -0.5 <i>Sharp deterioration</i>	-0.5 to -0.15 <i>Weakening</i>	-0.15 to +0.15 <i>Stable</i>	+0.15 to +0.5 <i>Acceleration</i>	> +0.5 <i>Strong acceleration</i>
Yearly	< -0.2 <i>Sharp deterioration</i>	-0.2 to -0.1 <i>Weakening</i>	-0.1 to +0.1 <i>Stable</i>	+0.1 to +0.2 <i>Acceleration</i>	> +0.2 <i>Strong acceleration</i>

This approach ensures that comparisons remain coherent with the dataset's temporal structure (daily, weekly, or monthly) and that the signal automatically adapts to sample length without manual tuning.

Highly visual and intuitive, it immediately shows whether an asset is accelerating or decelerating. It also validates momentum consistency: if momentum is positive but ΔReturn is negative, the trend is weakening.

WARNING — Sensitive to short-lived volatility spikes: a single event may artificially inflate short-term variation. The indicator also ignores persistence (a sharp rise followed by a sharp drop may cancel out the signal). Its relevance depends on historical depth, as if the LT window is too short, the contrast loses meaning.

5.2 Momentum Ratio

The Momentum Ratio compares current momentum strength to that observed over a longer period. It is a relative indicator showing whether recent dynamics are accelerating, stabilizing, or weakening.

Momentum measures the speed of price movement, the pace at which an asset rises or falls. The Momentum Ratio therefore expresses the relationship between present and past speed, highlighting whether motion is intensifying or fading.

- A ratio > 1 indicates accelerating momentum, the recent trend is stronger than the long-term one.
- A ratio < 1 indicates deceleration, the current dynamic is weaker than the historical baseline.

The formula used in the program is:

$$\text{Momentum Ratio} = \text{Momentum } \{CT\} / \text{Momentum } \{LT\}$$

Both momentum values (short- and long-term) are computed from dynamic rolling windows automatically adjusted to detected frequency.

Momentum ratio interpretation

Ratio Value	Interpretation	Context Description
< 0.5	Momentum collapse	The trend is reversing or has completely lost its strength.
0.5 – 0.8	Fading momentum	The move is losing steam — trend becomes fragile or uncertain.
0.8 – 1.2	Stable momentum	Momentum remains consistent with the long-term trend.
1.2 – 1.5	Strengthened momentum	Acceleration of the move — sign of consolidation or recovery.
> 1.5	Overpowered momentum	Rapid acceleration, often associated with a rally or short squeeze.

This is a robust, universal indicator that works across all asset classes and frequencies. It reveals divergences: if the ratio collapses while return remains positive, the trend is losing strength.

Although it relies on the same variables as Signal Stability (short- and long-term momentum), its purpose differs: the Momentum Ratio measures *relative intensity* of motion, whereas Signal Stability evaluates *directional coherence* across horizons. The two are therefore complementary: one describes speed, the other consistency.

WARNING — Strongly influenced by momentum volatility itself: brief spikes may distort the ratio. If long-term momentum is flat (near zero), the ratio becomes unstable. It also provides no directional information (bullish vs bearish); it only measures relative strength.

5.3 Volatility Ratio

The Volatility Ratio measures the relative change in short-term versus long-term volatility levels. It is a gauge of market-state transition, showing whether the environment is becoming more unstable or calming down.

Volatility reflects the average amplitude of price movements. Comparing short-term volatility (V_{CT}) with long-term volatility (V_{LT}) reveals regime shifts:

- When the ratio rises, the market grows more nervous and reactive.
- When it falls, the market re-enters a calm or compressed phase.

This indicator is central to identifying the four regimes used elsewhere in the report: subdued → normal → elevated → turbulent. The program uses the following formula:

$$\text{Volatility Ratio} = \text{Volatility } \{CT\} / \text{Volatility } \{LT\}$$

Volatility ratio interpretation

Ratio	Interpretation	Context Description
< 0.5	Volatility collapsing	Sharp contraction in risk — calm market or volatility compression.
0.5 – 0.8	Volatility compressing	Gradual reduction in dispersion — often a precursor to a breakout.
0.8 – 1.2	Volatility normalizing	Stable volatility — balanced market conditions.
1.2 – 1.5	Volatility expanding	Increasing dispersion — signs of tension or moderate stress.
> 1.5	Volatility spiking	Strong expansion — panic, turbulence, or macro shock.

It is a structural indicator of the market's risk regime and forms the basis of the automatic classification (subdued, normal, elevated, turbulent). It can also anticipate transitions before they appear in prices (rising volatility → incoming stress).

WARNING — Measures only relative change, not absolute level: a ratio > 1 on a low-volatility asset may be insignificant. It can be biased on short histories (if LT volatility is underestimated). On illiquid assets, short-term volatility can behave erratically and distort the ratio.

5.4 Beta (β)

Beta (β) measures an asset's sensitivity to a reference index (usually the overall market). It is a directionally weighted correlation indicator expressing how much an asset amplifies or dampens market movements.

Beta answers a simple question:

"If the market moves 1 %, by how much does my asset move on average?"

- $\beta > 1$ → the asset amplifies market moves: more volatile and reactive to global fluctuations;
- $\beta < 1$ → the asset moves less than the market: defensive or weakly correlated;
- $\beta < 0$ → inverse correlation: the asset often moves opposite to the market (rare; typical of hedges or some commodities).

There are two ways to obtain this indicator:

- Beta is not always computed directly by the program but can be retrieved from *vol_profiles.parquet*, which itself sources it from Yahoo Finance.
- When Yahoo does not provide a beta (e.g. for recent assets, exotic ETFs, or non-US listings), the program computes it from price data in *sectors.parquet*.

When no external value is available, the program applies the standard statistical formula directly to normalized returns of the asset and its chosen benchmark (commonly S&P 500 or MSCI World):

$$\beta = \text{Cov}(R_{\text{asset}}, R_{\text{benchmark}}) / \text{Var}(R_{\text{benchmark}})$$

where:

- R_{asset} and $R_{\text{benchmark}}$ are logarithmic returns over the same period,
- both series are time-aligned,
- covariance and variance are computed on the long-term dynamic window (e.g. *window_vol_L*).

This ensures beta values consistent with actual data depth and comparable to those retrieved from Yahoo.

Beta interpretation

β Value	Label Used in Report	Interpretation
> 1.5	<i>very_high-beta</i>	Highly market-sensitive asset; amplifies movements in both directions.
1.2 – 1.5	<i>high-beta</i>	Offensive asset; fully participates in bull cycles but more vulnerable during corrections.
0.8 – 1.2	<i>neutral</i>	Moves broadly in line with the market, with no major bias.
0.6 – 0.8	<i>low-beta</i>	Moderately defensive asset; reacts partially to broad market swings.
< 0.6	<i>defensive</i>	Protective asset with low sensitivity to economic cycles.
< 0	<i>inverse</i>	Negatively correlated with the market; can serve as a potential hedge.

Beta is a universally understood, cross-asset indicator. It not only helps weight systemic risk and refine Risk-Adjusted Return interpretation, but also highlights structural portfolio biases (defensive, aggressive, neutral).

WARNING — Beta depends heavily on both calculation period and chosen benchmark: the same asset may display different betas against different indices. In equity markets, it does not capture idiosyncratic risk (firm-specific factors). During crises, correlations rise mechanically and betas tend to converge toward 1, reducing discriminative power.

6. Short-Term Scenarios

6.1 Objectives

Short-Term (CT) Scenarios were introduced to provide a synthetic and narrative reading of the market state, beyond raw figures. They translate into analytical language the dynamic combination between return, momentum, and volatility, identifying typical market phases (e.g. “stressed rally,” “capitulation,” “technical rebound,” etc.).

Each scenario is designed to make the diagnosis visual and intuitive while remaining objectively data-driven. The approach serves two main purposes:

- to provide an economic and behavioral context for the current position of the asset or portfolio;
- to help prioritize observations (high impact, moderate impact, light impact) according to risk level and movement robustness.

6.2 Operating Principle

Scenarios are automatically determined from three key dimensions:

1. Return (*R_label*): direction and intensity of average performance;
2. Momentum (*M_label*): pace and consistency of the trend;
3. Volatility Regime (*V_regime*): market risk state (subdued → turbulent).

The engine cross-analyzes these three axes through a hierarchical set of rules. The first condition met defines the primary scenario, while the impact level (light, moderate, high) is then adjusted based on additional risk factors:

- Liquidity (*Liquidity_Label*)
- Leverage (*Leverage_Label*)
- Market sensitivity (*Beta_Label*)

The strength of the scenario framework lies in its ability to interpret figures (return, volatility, momentum) through concrete market behaviors, offering a unified analytical lens across very different asset types (equities, ETFs, commodities, rates, etc.), thereby helping to identify critical phases (stress, capitulation) or confirm bullish cycles (uptrend, volatility compression).

6.3 Classification Logic for Impact Level

- Thin / illiquid liquidity → tends to downgrade one level (high → moderate)
- Presence of leverage → tends to increase risk (light → moderate, moderate → high)
- Very high Beta_Label → may raise risk (moderate → high); defensive → may lower it (high → moderate)

6.4 Possible Values Among Variables

- *R_label* (Short-term return) ∈ {strongly negative, negative, neutral, moderately positive, strong, very strong}
- *M_label* (Momentum) ∈ {bearish, neutral, bullish, accelerating, accelerating+}
- *V_regime* (Volatility regime) ∈ {subdued, normal, elevated, turbulent} (*Labels depend on dynamic windows and detected data frequency.*)

7. Scenario Catalogue

1. **Capitulation** — [high]

Trigger: R_label = strongly negative and V_regime = turbulent and M_label contains bearish.

Definition: Panic selling with extreme volatility; order books thin out rapidly as liquidity evaporates.

Watch for:

- Volatility Ratio sharply rising; Δ Return deeply negative
- Signal_Stability often showing reversal (LT/CT divergence)
- Actual liquidity (ADV10USD) to avoid execution traps

2. **Stress** — [high]

Trigger: R_label $\in$ {strongly negative, negative} and V_regime $\in$ {elevated, turbulent}.

Definition: Significant damage but not full panic; potential for continuation.

Watch for:

- Momentum Ratio < 1 (weakening)
- Volatility Ratio > 1.2 (expansion)
- Reactions on key supports / newsflow sensitivity

3. **Momentum squeeze** — [high]

Trigger: R_label = very strong and M_label $\in$ {accelerating, accelerating+} and V_regime $\in$ {elevated, turbulent}.

Definition: Explosive momentum-driven rally under high volatility; prone to whipsaws and short-covering.

Watch for:

- Elevated Volatility Ratio; possible short squeezes
- Signal_Stability consistent but fragile to reversals
- Ensure sufficient liquidity for entries/exits

4. **Rally under tension** — [high]

Trigger: R_label $\in$ {strong, very strong} and V_regime $\in$ {elevated, turbulent}.

Definition: Strong rally with underlying nervousness; volatility threatens trend durability.

Watch for:

- Momentum Ratio ≥ 1 (otherwise “fatigued rally”)
- Sharp pullbacks possible → apply strict risk control

5. **Regular uptrend** — [moderate]

Trigger: R_label $\in$ {strong, very strong} and M_label $\in$ {bullish, accelerating, accelerating+} and V_regime $\in$ {subdued, normal}.

Definition: Healthy bullish trend: gains supported by positive momentum and contained volatility.

Watch for:

- Signal_Stability ideally strong and coherent
- Volatility Ratio ≈ 1 (no overheating)

6. **Loss of momentum** — [moderate]

Trigger: M_label contains bearish and R_label $\notin$ {negative, strongly negative} and V_regime $\in$ {normal, elevated}.

Definition: Gradual loss of traction; potential consolidation or sector rotation phase.

Watch for:

- Momentum Ratio < 1; declining Δ Return
- Possible price/momentum divergences

7. **Technical rebound** — [moderate]

Trigger: M_label $\in$ {bullish, accelerating, accelerating+} and R_label $\in$ {negative, neutral}.

Definition: Technical rebound without confirmed returns; test phase following a drop.

Watch for:

- Signal_Stability (is CT/LT coherence improving?)
- R_label needs to turn positive for full confirmation

8. **Gradual decline** — [moderate]

Trigger: R_label = negative and V_regime $\in$ {subdued, normal} and M_label contains bearish.

Definition: Controlled decline without panic; sustained downward pressure.

Watch for:

- Volatility Ratio ≤ 1.0 (orderly selloff)
- Progressive break of support levels

9. **Volatility compression** — [light]

Trigger: V_regime = subdued and M_label $\in$ {neutral, bullish} and R_label $\in$ {neutral, moderately positive}.

Definition: Volatility compression; potential breakout setup.

Watch for:

- Range break; improving Signal_Stability
- Post-break jump in Volatility Ratio (volatility may surge right after a key breakout)

10. **Volatility expansion** — [moderate]

Trigger: V_regime $\in$ {elevated, turbulent} and $|\text{AvgReturn}(\%)| < 0.2$ and M_label = neutral.

Definition: Expanding volatility without directional bias (choppy market); frequent false signals.

Watch for:

- Wait for directional bias via R_label / M_label
- Avoid overtrading

11. **Stabilization after shock** — [moderate]

Trigger: V_regime = elevated and R_label = neutral and M_label ∈ {neutral, bullish}.

Definition: Decrease in nervousness after a shock; breathing phase.

Watch for:

- Volatility Ratio reverting toward 1
- Confirmation via ΔReturn turning positive again

12. **Distribution** — [moderate]

Trigger: R_label ∈ {strong, moderately positive} and M_label contains bearish and V_regime ∈ {normal, elevated}.

Definition: Profit-taking / trend fatigue; risk of rotation.

Watch for:

- Momentum Ratio < 1; emerging divergences
- R_label shifting from positive to neutral, then negative

13. **Range / noise** — [light]

Trigger: (no previous rule applies).

Definition: Sideways / noisy market; alternating micro-movements without a clear trend.

Watch for:

- Volatility Ratio ≈ 1; Signal_Stability often neutral
- Suitable strategies: mean reversion, stock-picking

The thirteen CT scenarios defined in the engine form a complete and balanced mapping of market regimes. They cover all relevant configurations derived from the combination of Return, Momentum, and Volatility Regime from panic phases (capitulation, stress) to recovery or neutral phases (uptrend, range).

This number was deliberately chosen to balance analytical precision and readability: each scenario remains distinct and interpretable, without redundancy or excessive complexity. Beyond this threshold, additional cases would only add false granularity at the expense of clarity and overall diagnostic coherence.

8. Visualizations & Limitations

8.1 Number of tickers

The report dynamically adapts to the number of selected assets: all visualizations, analyses, and interpretations are automatically generated without any predefined technical limit. However, for readability and graphical coherence, it is recommended to limit the number of tickers to 10–20 for a focused sectoral analysis, and to 40–50 for a full portfolio review. Beyond these thresholds, visual clarity and interpretability may degrade, although report generation remains fully functional.

8.2 Recommended observation window sizes

Dynamic windows automatically adjust to the chosen frequency (daily, weekly, monthly, yearly). However, to ensure statistical stability of key indicators (Return, Momentum, Volatility), certain bounds are recommended:

Recommended bounds by analysis frequency

Analysis Frequency	Recommended Minimum Window	Recommended Maximum Window	Rationale
Daily	15 to 25 observations	60 to 100 observations	Below 15 days, noise dominates the trend; beyond 100, responsiveness decreases.
Weekly	4 to 8 observations	20 to 30 observations	Captures intermediate cycles without over-smoothing signals.
Monthly	3 to 6 observations	12 to 18 observations	Sufficient to detect macro regime shifts without diluting dynamics.
Yearly	2 to 3 observations	5 to 8 observations	Intended for long-term structural analyses; beyond this range, responsiveness becomes too low.

8.3 Temporal resolution and data quality

All prices used in the report come from Yahoo Finance, a reliable and widely recognized source for large-cap equities and liquid ETFs. For these assets, price adjustments (dividends, splits, closing variations) are considered accurate and fully suitable for quantitative analysis.

However, for small-cap stocks (below USD 2 billion) or for recent and thinly traded assets, occasional irregularities may appear (missing values, delayed adjustments, or offset timestamps). These anomalies do not affect the structure of the report but can slightly distort rolling indicators such as volatility, short-term momentum, or Δ Return.

Recommendation: perform a quick visual check of price series for any asset with a market capitalization below USD 2 billion before including it in the report.

8.4 Frequency bias and aggregation

Indicators are recalculated separately for each frequency (Daily, Weekly, Monthly, Yearly). → A frequency change mechanically alters returns, volatility, and correlations because observation windows and time granularity are no longer comparable.

Recommendation: avoid directly comparing reports of different frequencies without prior annualization of indicators, in order to normalize all measures to a common time base.

Annualization formulas by frequency

Frequency	Annualized Return	Annualized Volatility
Daily	$R_{ann} = R_{daily} \times 252$	$\sigma_{ann} = \sigma_{daily} \times \sqrt{252}$
Weekly	$R_{ann} = R_{weekly} \times 52$	$\sigma_{ann} = \sigma_{weekly} \times \sqrt{52}$
Monthly	$R_{ann} = R_{monthly} \times 12$	$\sigma_{ann} = \sigma_{monthly} \times \sqrt{12}$
Yearly	— (already annualized)	— (already annualized)

8.5 Sensitivity to outliers

Some indicators (such as Momentum or ΔReturn) can be heavily influenced by short-term spikes or exceptional variations. → This may distort scenario perception if an unusual event (earnings release, major news, or crisis) occurs within the analysis window.

Recommendation: interpret recent results with caution and always cross-check with other time horizons.

9. Notes and Clarifications

9.1 Scaling factors by asset class (*)

This table provides the scaling coefficients applied to dimensioned indicators (Return, Volatility, Momentum, Δ Return) depending on the asset class. Its goal is to ensure a consistent and comparable interpretation across heterogeneous assets: a ± 1 % move does not carry the same economic weight for a currency pair as for a commodity or an equity index. These multiplicative factors normalize each measure to the “equity baseline” (factor = 1.0), which serves as the interpretive reference throughout the report.

Indicators affected:

- Return (%)
- Volatility (%)
- Momentum (%)
- Δ Return (pp)

Asset class	Return	Volatility	Momentum	Summary comment
Currencies (FX)	× 0.3	× 0.5	× 0.7	Low amplitude and structurally weak returns; highly mean-reverting market.
Bonds (Fixed Income)	× 0.5	× 0.6	× 0.8	Stable cash flows, limited volatility; increased sensitivity to interest rates.
Equities / ETFs / Indices	× 1.0	× 1.0	× 1.0	Reference asset class; balanced dynamics between risk and return.
Commodities	× 1.2	× 1.3	× 1.3	Cyclical assets sensitive to macro cycles and supply-demand shocks.
Listed Real Estate (REITs)	× 0.9	× 1.1	× 1.0	Partially rate-sensitive; tends to smooth trends over time.
Cryptocurrencies (Crypto)	× 2.5	× 2.0	× 1.8	Extremely volatile; directional moves are sharp and discontinuous.
Alternative assets (PE, Hedge Funds)	× 0.8	× 0.8	× 1.0	Smoothed returns; volatility often underestimated due to illiquidity.

WARNING — The scaling table below should not be applied to leveraged products (e.g. ETF x2, x3, etc.) or to derivatives whose structural leverage amplifies both return and volatility. In such cases, the leverage factor takes precedence over the asset class scaling. For example, a 3x leveraged Nasdaq ETF should retain the “Equity = 1.0” scaling baseline but have its Return and Volatility values multiplied by 3 to reflect the true exposure.

9.2 Glossary of technical terms

This glossary compiles the main technical terms and expressions used throughout the report. It aims to clarify the financial and behavioral concepts underlying market dynamics, making the indicators easier to read and interpret.

Portfolio analysis	Comprehensive study of a set of assets aimed at evaluating diversification, correlation, and overall risk profile.
Targeted sector analysis	Analysis limited to a group of assets belonging to the same economic sector (e.g., technology, energy) in order to assess internal rotation and performance consistency.
Choppy	A term describing an irregular market with no clear direction, marked by fast and contradictory price movements.
Price/momentum divergence	A situation where an asset's price continues to move in one direction while momentum (speed of movement) weakens or reverses; often a sign of a potential trend reversal.
Short-covering effect	Rapid price increase caused by the simultaneous closing of short positions, forcing sellers to buy back shares to limit their losses.
CSV file	Structured text file containing tabular data separated by commas. Used here to store descriptive metadata about assets (symbol, sector, etc.). In this case, the file can be edited directly.

Parquet file	Binary file format optimized for fast storage and reading of large data tables. Used for historical price series and volatility profiles. Must be edited through pandas or an equivalent tool.
Benchmark index	Reference index used to compare the performance of an asset or portfolio. Examples: S&P 500 for U.S. equities, MSCI World for global allocation, Bloomberg Barclays Aggregate for bonds, or the CRB Index for commodities.
Operational liquidity of a security	An asset's ability to absorb large trading volumes without causing excessive price variations. Measured by the ADV10USD indicator.
Mean reversion	Strategy based on the assumption that prices tend to revert to their mean after excessive upward or downward moves. Opposite of trend-following strategies.
Newsflow	Stream of news and announcements (earnings, macro releases, policy decisions, etc.) likely to influence short-term prices.
Overtrading	Excessive trading activity without fundamental justification, often driven by high volatility or loss of discipline.
Volatility spikes	Sudden and temporary surges in volatility, often triggered by exogenous events (earnings releases, crises, unexpected announcements).
Pullback	Temporary price move back toward a former support or resistance level after a breakout, often interpreted as a test of the initial move.
Sharpe ratio	Classic measure of risk-adjusted performance: (portfolio return – risk-free rate) / volatility. The higher it is, the more efficient the return.
Technical rebound	Temporary upward movement following a sharp decline, without any fundamental trend change. Often driven by short-term overselling or technical adjustments.
Reaction on key supports	Price behavior around major technical or psychological levels (supports or resistances), often used to detect potential reversals.
Breakout risk	Risk that a price breakout beyond a key level fails or reverses quickly, producing a false signal and a sharp return to the previous range.
Consolidation/rotation risk	Phase during which gains stabilize or rotate between sectors. May precede a rebound, correction, or change in leadership.
Imminent reversal risk	Situation where multiple indicators (momentum, volatility, signal stability) converge to signal a high probability of short-term trend reversal.
Idiosyncratic risks	Risks specific to an individual asset or company, independent of the overall market (e.g., governance issue, unexpected results, legal event).
Support break	Breakdown below a key technical level, often interpreted as a sign of weakness and continuation of the downward trend.
GICS sectors	Global Industry Classification Standard: categorization of companies into 11 main sectors (energy, finance, healthcare, etc.).
CT scenario	Short-term market configuration identified by the report engine (e.g., Capitulation, Stress, Rally under tension). Based on the combination of Return / Momentum / Volatility.
Breakout setup	Technical configuration indicating a likely breakout from a range or resistance, often preceded by volatility compression.
Squeeze	Phase where short sellers are forced to buy back massively, triggering a fast and self-reinforcing price increase.
Stock picking	Selection of individual stocks based on specific criteria (value, growth, momentum), as opposed to index-based investing.
Whipsaws	Rapid back-and-forth price swings invalidating multiple technical signals in a row, typical of sideways or unstable markets.